Face Recognition with Logistic Regression and PCA

Identifying faces from the Olivetti dataset using scikit-learn and PyTorch logistic regression, with PCA-based dimensionality reduction.

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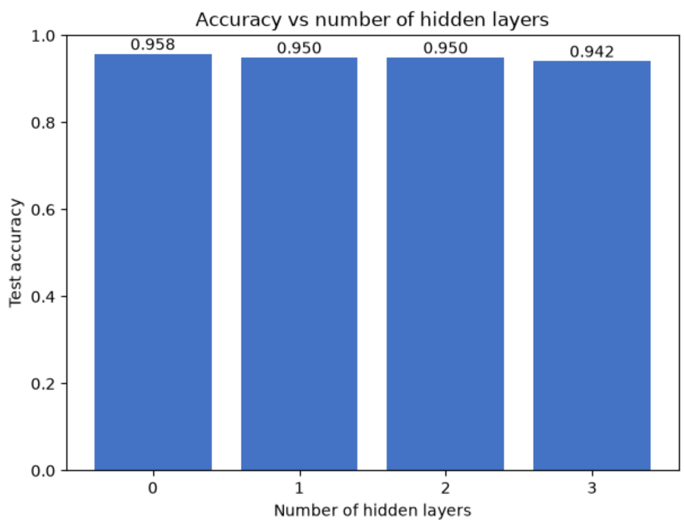
Overview

Face recognition is a classic multi-class classification problem that tests a model's ability to distinguish fine-grained visual patterns. This project applies Logistic Regression to the Olivetti faces dataset (400 grayscale images of 40 individuals, each represented as a 4096-dimensional vector) using two frameworks: scikit-learn and PyTorch. PCA is used to explore how dimensionality reduction affects recognition accuracy.



Recognition accuracy across the compared models.

Goals

Build and evaluate logistic regression classifiers for face recognition, compare scikit-learn and PyTorch implementations, and analyse how architecture choices (hidden layers, optimizer, PCA components) affect accuracy on a held-out test set.

Objectives

- Load and prepare the Olivetti faces dataset with a 70/30 stratified train/test split.
- Train a scikit-learn Logistic Regression model with 10-fold cross-validation and evaluate on the test set.
- Implement an equivalent model in PyTorch as a single linear layer (4096 to 40), and compare performance to scikit-learn.
- Analyse accuracy vs the number of hidden layers and vs the choice of optimizer (SGD, Adam, RMSprop, Adagrad).
- Apply PCA and plot Logistic Regression accuracy as the number of principal components increases

from 2 to 10.

Approach

Data is loaded via sklearn's fetch_olivetti_faces and split into training and test sets. The scikit-learn pipeline uses KFold cross-validation to estimate generalisation performance before final test evaluation. The PyTorch pipeline standardises features with StandardScaler (fit on the training set only), then trains models with varying architectures and optimizers. PCA experiments reduce the feature space before fitting a logistic regression model, measuring accuracy at each component count.

Outcomes

- Scikit-learn Logistic Regression achieved 10-fold CV accuracy of 0.939 and test accuracy of 0.975.
- PyTorch Logistic Regression achieved 10-fold CV accuracy of 0.932 and test accuracy of 0.958.
- Adding hidden layers (up to 3 layers) did not improve test accuracy; the plain linear model performed best on this small dataset.
- PCA accuracy on the test set increased from 0.15 with 2 components to 0.88 with 10 components, showing strong dependence on the number of retained dimensions.